

- 7 (a) A chloroplast is composed of many structures, each with a different function. Several chloroplast structures are listed.

stroma lamellae thylakoid membrane ribosome
thylakoid space starch grain DNA outer membrane

From the list:

- (i) identify the structures involved in the production of rubisco

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[2]

- (ii) identify the structure that contains a high concentration of protons in daylight.

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[1]

- (b) Paper chromatography is a technique that can be used to separate a mixture of four common chloroplast pigments. The pigments can be identified by calculating their R_f values.

A student carried out paper chromatography on a solution containing a mixture of chloroplast pigments.

The results are shown in Table 7.1.

Table 7.1

pigment	distance travelled by pigment from baseline / cm	distance travelled by solvent from baseline / cm	R_f
.....	6.5	8.9	0.73
chlorophyll a	4.6	8.9	0.52
chlorophyll b		8.9	0.38
carotene	8.2	8.9	

Complete Table 7.1.

[3]

(c) Fig. 7.1 shows the absorption spectrum for carotene and for chlorophyll a.

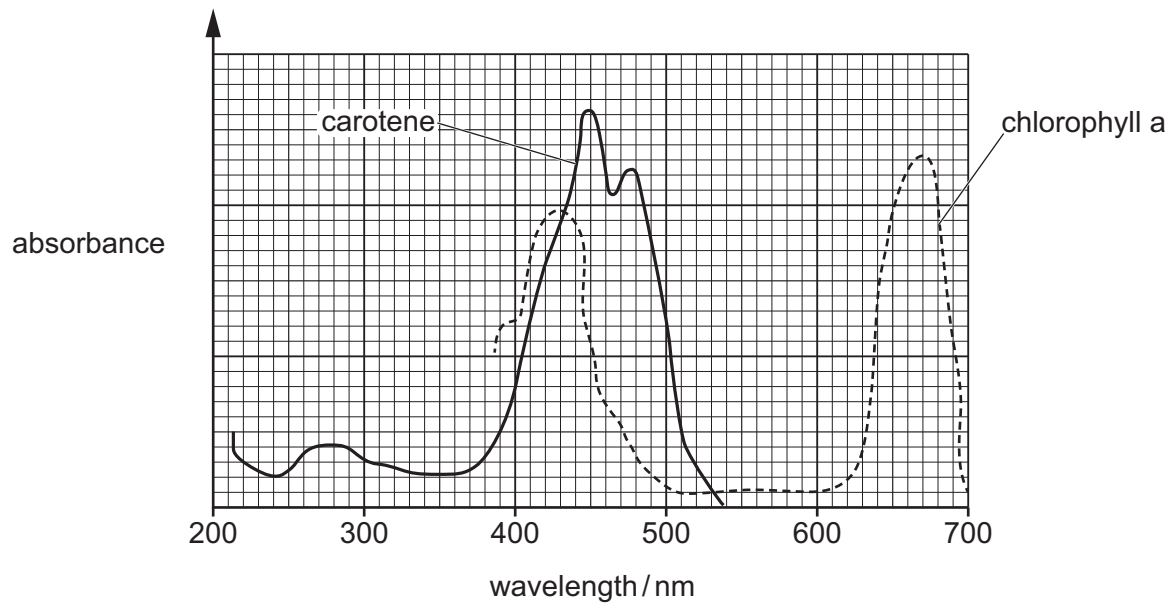


Fig. 7.1

With reference to Fig. 7.1, describe **and** explain the role of carotene in photosynthesis.

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..... [4]

[Total: 10]